This PDF was created with https://tianfei.chat , free download: https://tianfei.chat/article/one-click-webpage-to-pdf-tools/?from=saved_pdf_file
From: http://118.185.187.137/moodle/mod/quiz/review.php?attempt=369188&cmid=3322

5-G-Product of Array element...
Divide and Conquer
1-Number of Zeros in a Given ...
2-Majority Element
3-Finding Floor Value
4-Two Elements sum to x
5-Implementation of Quick Sort
Dynamic Programming
1-DP-Playing with Numbers
2-DP-Playing with chessboard
3-DP-Longest Common Subse...
4-DP-Longest non-decreasing ...
Competitive Programming
1-Finding Duplicates-O(n^2) Ti...
2-Finding Duplicates-O(n) Tim...
3-Print Intersection of 2 sorte...

1-DP-Playing with Numbers

Started on	Saturday, 12 September 2026, 2:27 PM
State	Finished
Completed on	Saturday, 12 September 2026, 2:31 PM
Time taken	3 mins 56 secs
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Correct Mark 10.00 out of 10.00 Flag question

Playing with Numbers:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer ‘n’ and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

Example 1:

Input: *6*
Output:*6*
Explanation: *There are 6 ways to 6 represent number with 1 and 3*

1+1+1+1+1+1
3+3
1+1+1+3
1+1+3+1
1+3+1+1
3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways ‘n’ can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    long long dp[n + 4];

    dp[0] = 1;
    dp[1] = 1;
    dp[2] = 1;

    for (int i = 3; i <= n; i++) {
        dp[i] = dp[i - 1] + dp[i - 3];
    }
    printf("%lld\n", dp[n]);

    return 0;
}
```

	Input	Expected	Got	
✔	6	6	6	✔
✔	25	8641	8641	✔
✔	100	24382819596721629	24382819596721629	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Finish review

Back to Course

WEBxPDF
Converting your page to PDF...